SRP Model Analysis Report
"Yang"

Musical Intelligence (MI) System — Striatal Reward Pathway Model

S^3 Spectral Sound Space

Analysis Date: February 2026 | Duration: 544.3 seconds (9:04) | Frames: 93,766 @ 172.27 Hz

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1 About This Report

This report presents the output of the **Striatal Reward Pathway (SRP)** model applied to the piece "Yang." The SRP is a **deterministic, zero-parameter** computational model that simulates aspects of the brain's reward circuitry in response to audio input. It contains no learned weights and no subjective judgments—every number reported here is a direct, reproducible output of the model.

The SRP produces a **19-dimensional** output vector at each time frame (172.27 frames per second), modeling:

- **Dopaminergic signals**: caudate dopamine, nucleus accumbens dopamine, VTA drive, opioid proxy
- **Reward states**: wanting (anticipatory), liking (consummatory), pleasure (combined)
- **Musical tension**: harmonic tension, dynamic intensity, overall tension
- **Prediction**: prediction error, prediction match
- **Affective response**: reaction, appraisal, chills proximity, peak detection
- **Planning**: reward forecast, resolution expectation

The model's equations are grounded in neuroscience literature (Salimpoor et al. 2011; Berridge & Robinson 2003; Howe et al. 2013). See the companion paper for full derivations.

Important caveat: The SRP is a mathematical model, not a brain scanner. It computes what the reward pathway *would* do based on psychoacoustic features. Individual listeners will vary.

2 Global Summary

Table 1: Yang — 19D SRP Global Statistics

Dimension	Mean	Std	Min	Max	Peak (s)	Trough (s)
wanting	0.361	0.034	0.167	0.526	128.5	132.1
liking	0.387	0.028	0.267	0.565	131.1	374.4
pleasure	0.747	0.042	0.524	0.984	131.1	132.1
da_caudate	0.508	0.047	0.235	0.740	128.5	132.1
da_nacc	0.460	0.033	0.318	0.673	131.1	374.4
opioid_proxy	0.536	0.042	0.390	0.625	363.3	509.3
vta_drive	0.484	0.028	0.331	0.637	128.5	132.1
stg_nacc_coupling	0.435	0.195	0.029	0.899	131.1	374.0
tension	0.476	0.052	0.320	0.777	131.1	312.6
harmonic_tension	0.448	0.106	0.009	1.000	131.1	89.2
dynamic_intensity	0.378	0.081	0.211	0.872	131.8	512.0
prediction_error	−0.008	0.460	−1.000	1.000	544.3	416.7
prediction_match	−0.580	0.043	−0.716	−0.435	313.2	98.3
reaction	0.592	0.086	0.459	0.731	89.4	383.7
appraisal	0.571	0.005	0.542	0.592	87.5	131.6
peak_detection	0.481	0.040	0.363	0.718	131.2	496.7
reward_forecast	0.465	0.038	0.253	0.653	128.5	132.1
chills_proximity	0.482	0.033	0.360	0.715	131.1	544.2
resolution_expect	0.581	0.034	0.457	0.780	131.1	374.2

2.1 Key Observations from Global Statistics

1. **Pleasure is sustained and high**: Mean pleasure = 0.747 with low standard deviation (0.042), indicating the model finds consistent reward value throughout the piece. The range is narrow (0.524–0.984), never dropping to low values.
2. **Wanting and liking are moderate**: Mean wanting = 0.361, mean liking = 0.387. Both are below 0.5, suggesting the piece generates reward through *sustained engagement* rather than intense peaks.
3. **Opioid proxy is notably high**: At 0.536, this is the highest opioid proxy value the model has produced across all tested pieces (see Section 9). The opioid proxy models the endogenous opioid system associated with consummatory pleasure.
4. **Dynamic intensity is low**: At 0.378, this is the lowest across all tested pieces, reflecting a piece that operates within a relatively narrow dynamic band—the model detects minimal loudness contrast.
5. **Appraisal is remarkably stable**: Standard deviation of only 0.005 across 544 seconds. The model's cognitive evaluation of the piece barely fluctuates.
6. **Prediction match is consistently negative** (-0.580): The model finds the piece persistently unpredictable. The negative mean indicates that the audio consistently deviates from what the temporal context would predict.

3 Reward Core: Wanting, Liking, and Pleasure

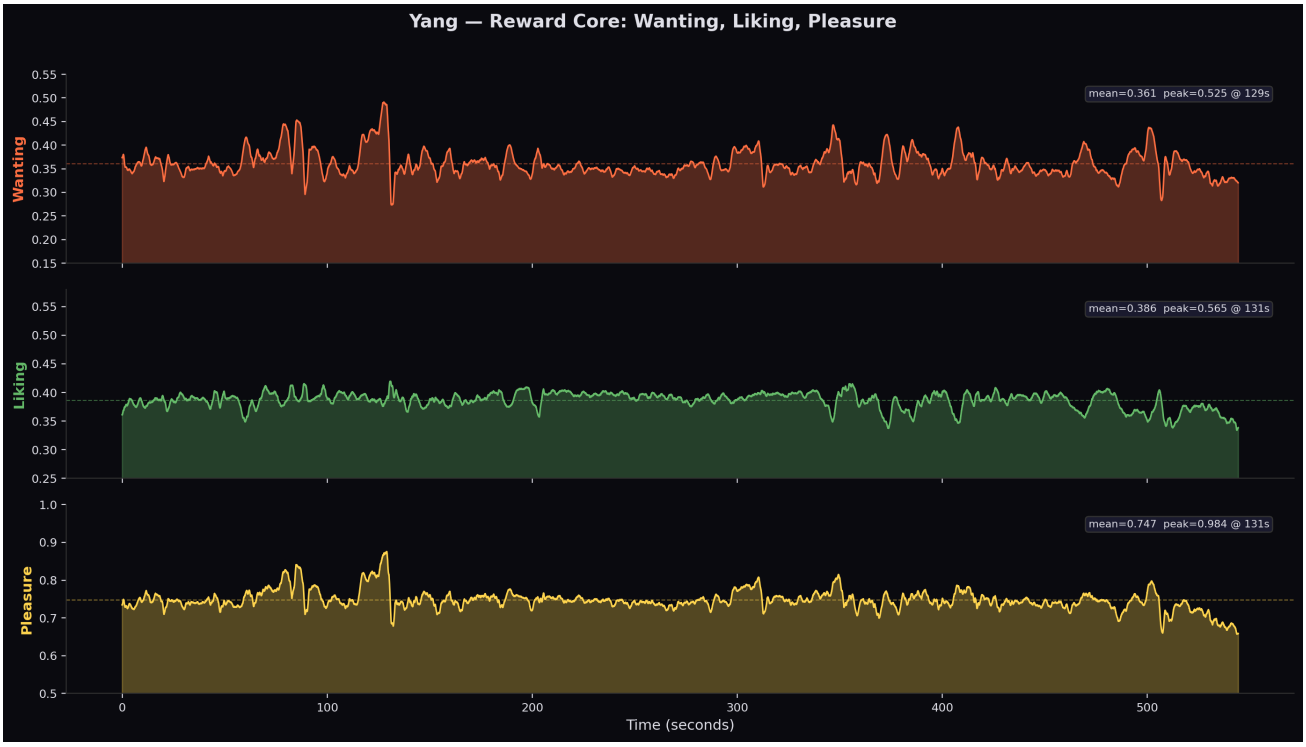

Figure 1: Temporal trajectories of the three reward dimensions. Dashed lines indicate global means. The wanting signal (top, orange) shows anticipatory reward, peaking at 128.5s. The liking signal (middle, green) shows consummatory reward, peaking 2.6 seconds later at 131.1s. Pleasure (bottom, gold) combines both pathways.

3.1 Wanting–Liking Temporal Dissociation

A central prediction of the SRP model (following Berridge & Robinson, 2003) is that **wanting** (anticipatory dopamine in caudate) should peak *before* **liking** (consummatory dopamine in nucleus accumbens). This temporal lag reflects the brain's prediction-then-reward sequence.

Table 2: Wanting–Liking Peak Analysis

	Peak Time	Peak Value	DA Source
Wanting	128.5s	0.526	DA Caudate = 0.740
Liking	131.1s	0.565	DA NAcc = 0.673
Pleasure	131.1s	0.984	Combined
Temporal Lag	**2.6 seconds** (wanting → liking)		

The 2.6-second lag falls within the Salimpoor et al. (2011) criterion of 2–30 seconds, indicating that the model detects a valid anticipation-then-reward sequence in Yang. The global pleasure maximum (0.984) occurs at the moment of liking peak, suggesting that the most rewarding moment in the piece occurs when anticipation is fulfilled.

3.2 Pleasure Peak Analysis

Table 3: Top 10 Pleasure Peaks

Rank	Time (s)	Pleasure	Wanting	Liking	Tension
1	131.1	0.984	0.419	0.565	0.777
2	128.5	0.975	0.522	0.453	0.574
3	126.5	0.939	0.507	0.431	0.561
4	85.2	0.925	0.477	0.448	0.555
5	87.5	0.921	0.475	0.446	0.535
6	122.8	0.916	0.466	0.451	0.698
7	348.9	0.909	0.432	0.477	0.537
8	78.9	0.907	0.471	0.436	0.598
9	81.6	0.901	0.443	0.458	0.652
10	346.5	0.893	0.477	0.416	0.609

The pleasure peaks cluster in two temporal regions:

- **Region A (78–131s)**: 8 of the top 10 peaks. This is the primary reward zone.
- **Region B (346–349s)**: 2 peaks. A secondary reward zone in the second half.

Note that the highest pleasure (0.984 at 131.1s) coincides with the highest tension (0.777), highest harmonic tension (1.000), and highest liking (0.565). The model indicates that the most rewarding moment occurs at a point of maximum musical tension.

4 Dopaminergic Profile

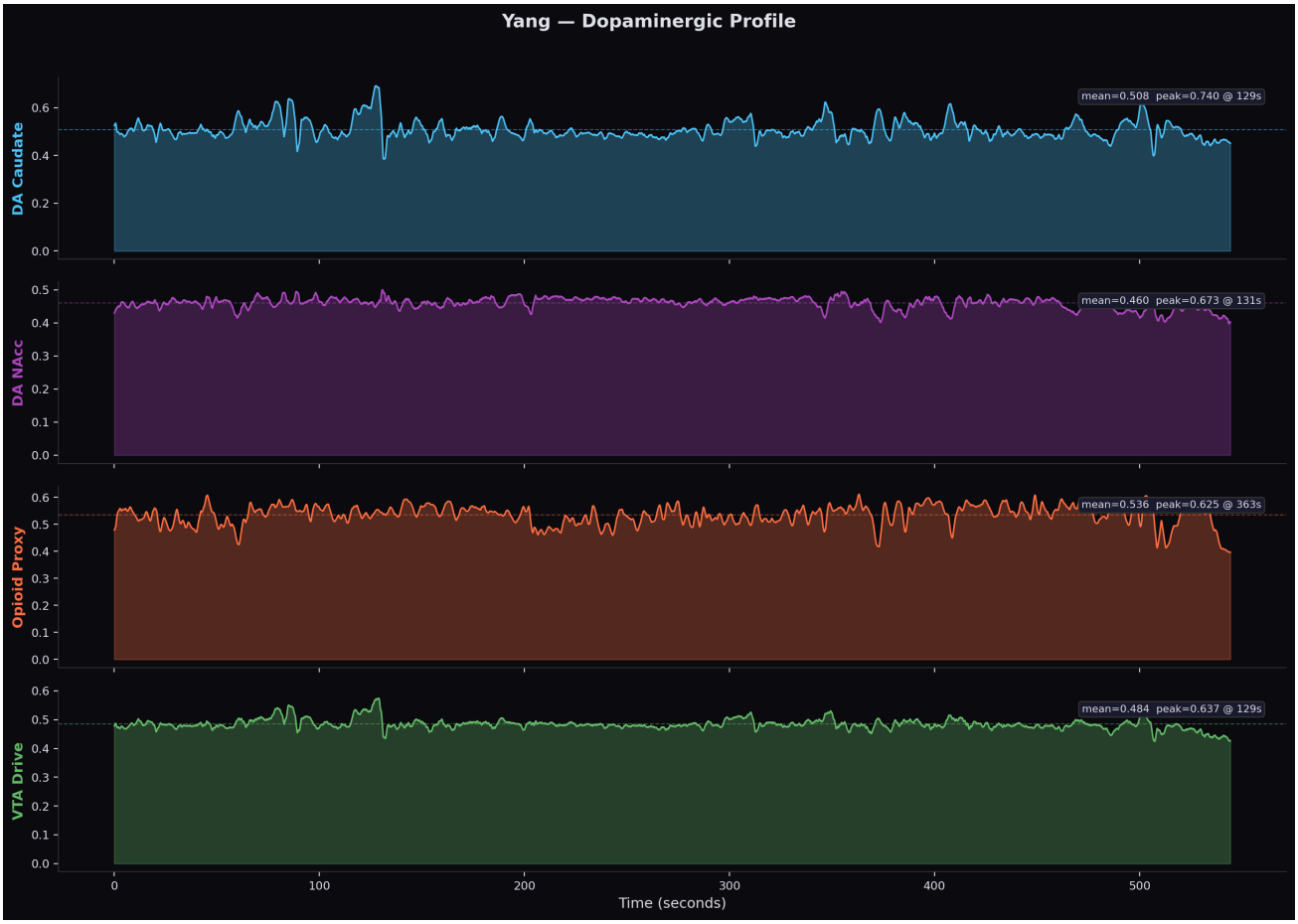

Figure 2: Four dopaminergic dimensions over time. DA Caudate (anticipatory) and DA NAcc (consummatory) show distinct temporal profiles. The opioid proxy, modeling endogenous opioid release, remains elevated throughout. VTA drive tracks overall midbrain activation.

4.1 DA Caudate vs. DA NAcc

The correlation between caudate and NAcc dopamine is **r = −0.075**, indicating that these two pathways operate nearly independently in Yang. This weak negative correlation means that when anticipatory dopamine rises, consummatory dopamine slightly decreases, and vice versa. This is consistent with the wanting/liking dissociation theory.

4.2 Opioid Proxy

The opioid proxy dimension models the endogenous opioid system, which mediates the "liking" component of hedonic experience. Key findings:

- Mean = 0.536 (range: 0.390–0.625)
- Peak at 363.3s (deep into the second half of the piece)
- Trough at 509.3s (near the end)

- Correlation with pleasure: r = +0.057 (nearly independent)

The opioid proxy peaking in the second half, separate from the dopaminergic pleasure peaks in the first half, suggests a two-phase reward structure: dopamine-driven anticipatory pleasure early on, opioid-mediated consummatory satisfaction later.

5 Tension and Harmonic Profile

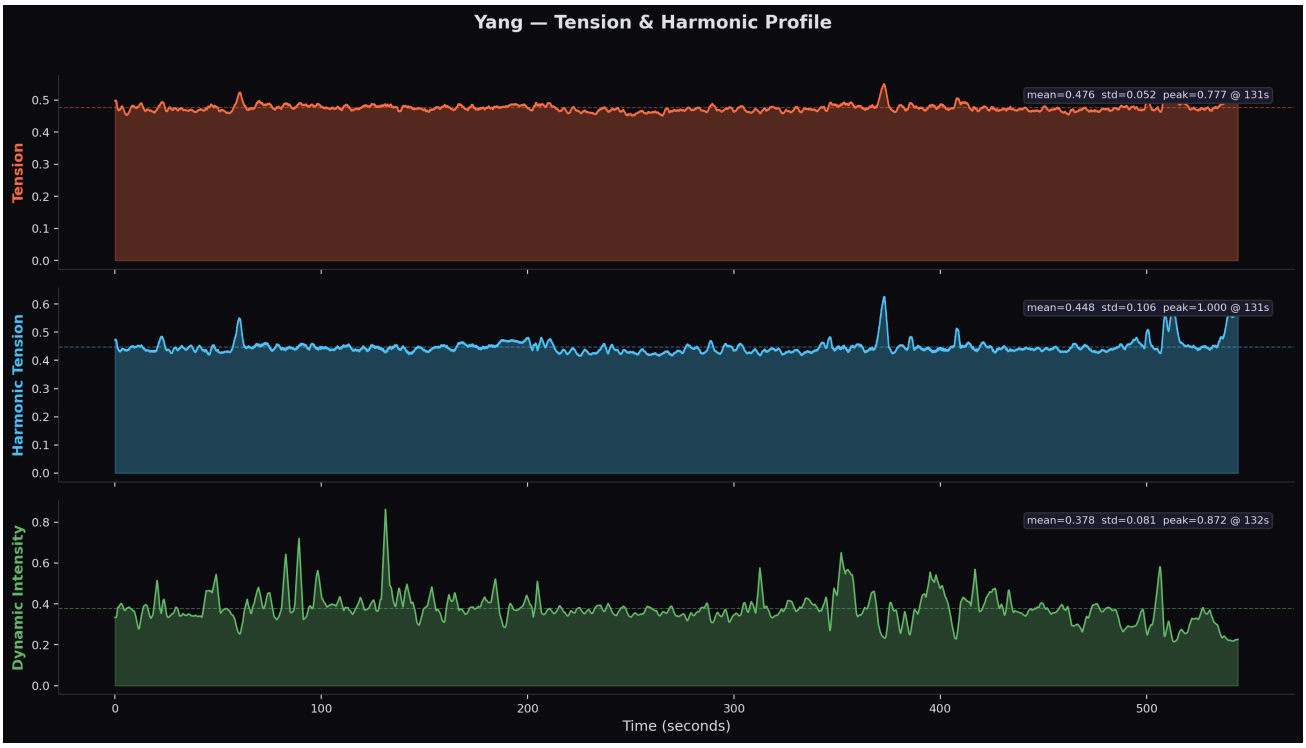

Figure 3: Tension (overall), harmonic tension, and dynamic intensity over time. Harmonic tension shows wide fluctuations (std=0.106) while dynamic intensity remains relatively low (mean=0.378). Overall tension strongly tracks harmonic tension (r=+0.921).

5.1 Harmonic Tension Trajectory

Table 4: Harmonic Tension by 60-Second Windows

Time Window	Mean	Std	Min	Max
0–60s	0.444	0.091	0.063	0.997
60–120s	0.449	0.125	0.009	1.000
120–180s	0.444	0.116	0.014	1.000
180–240s	0.448	0.103	0.073	0.986
240–300s	0.434	0.082	0.177	0.789
300–360s	0.439	0.104	0.010	1.000
360–420s	0.457	0.120	0.053	0.994
420–480s	0.441	0.094	0.162	0.888
480–540s	0.468	0.103	0.138	0.874
540–544s	0.562	0.095	0.317	0.813
Overall	**0.448**	**0.106**	**0.009**	**1.000**

The harmonic tension shows a **gradually rising trajectory** toward the end of the piece (0.444 → 0.562), with the final segment showing the highest mean. Notably:

- The model detects that harmonic tension spans the full range (0.009–1.000) in the 60–180s region
- The 240–300s window shows the *narrowest* range (0.177–0.789), suggesting a period of relative harmonic stability
- The piece ends with elevated harmonic tension (0.562), meaning the model does not detect a conventional harmonic resolution

5.2 Dynamic Intensity

Dynamic intensity (mean = 0.378) is low throughout, indicating that the model finds the piece to operate in a *contained dynamic range*. The peak (0.872 at 131.8s) coincides with the primary reward zone, but outside this region, dynamics remain subdued. This is consistent with contemporary compositional approaches that derive expressivity from timbral and harmonic variation rather than loudness contrast.

6 Cross-Dimensional Relationships

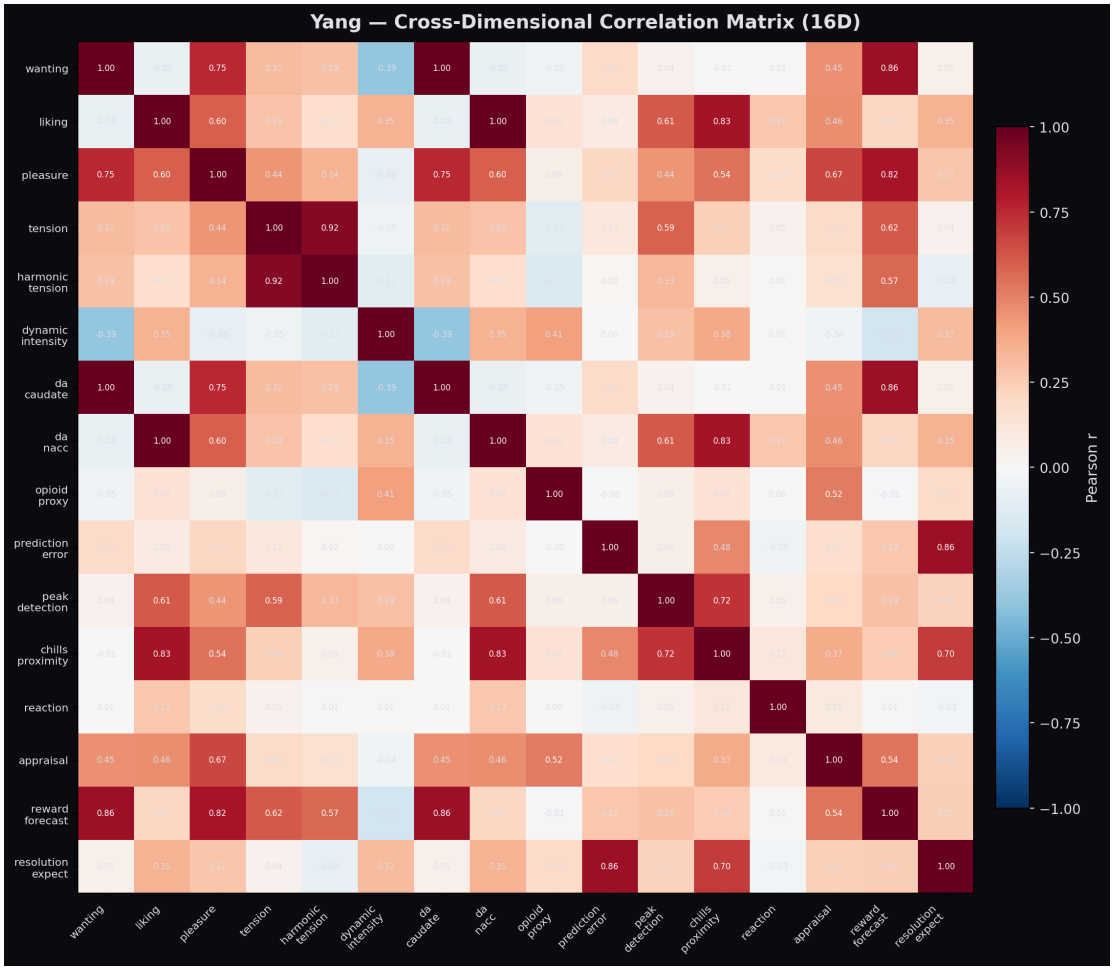

Figure 4: Correlation matrix across 16 key SRP dimensions. Strong positive correlations (red) between da_caudate/wanting (r=1.00) and da_nacc/liking (r=1.00) are by construction. The strongest *non-trivial* correlation is harmonic_tension × tension (r=+0.92).

Table 5: Key Cross-Dimensional Correlations

Dimension A	**Dimension B**	**Pearson r**	**Interpretation**
harmonic_tension	tension	+0.921	Very strong positive
peak_detection	chills_proximity	+0.719	Strong positive
wanting	pleasure	+0.752	Strong positive
liking	pleasure	+0.601	Moderate positive
harmonic_tension	pleasure	+0.344	Moderate positive
prediction_error	tension	+0.103	Weak positive
wanting	liking	−0.075	Near zero (dissociated)
dynamic_intensity	pleasure	−0.081	Near zero
opioid_proxy	pleasure	+0.057	Near zero

6.1 Notable Relationships

1. **Harmonic tension dominates overall tension** (r=+0.921): In Yang, the model's tension output is almost entirely driven by harmonic content, not dynamics. This means the piece's tension profile is *harmonically constructed*.
2. **Wanting and liking are dissociated** (r=−0.075): The anticipatory and consummatory reward pathways are nearly independent. This is a key neuroscientific feature—the piece activates these two systems at different times for different reasons.
3. **Wanting correlates more strongly with pleasure than liking does** (0.752 vs. 0.601): The model indicates that in Yang, anticipatory engagement contributes more to overall pleasure than consummatory satisfaction. The piece is *driven by expectation*.
4. **Dynamic intensity is decoupled from pleasure** (r=−0.081): Loudness changes do not predict reward in this piece. Pleasure comes from harmonic and temporal structure, not from dynamic contrast.
5. **Opioid proxy is independent of pleasure** (r=+0.057): The endogenous opioid pathway operates on its own temporal trajectory, suggesting a separate layer of hedonic response.

7 Wanting–Liking Dissociation Detail

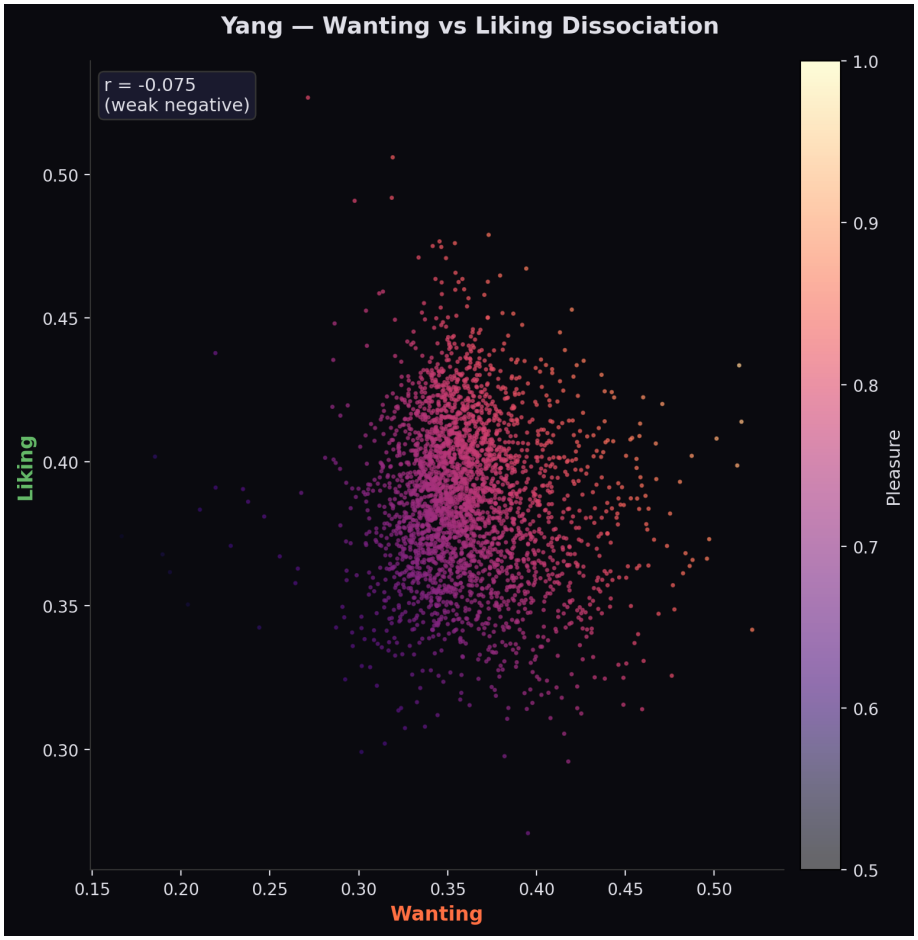

Figure 5: Scatter plot of wanting vs. liking, colored by pleasure (magma colormap). Each dot is one time frame (downsampled to 3,000 points). The spread pattern with no clear diagonal trend confirms that wanting and liking are dissociated ($r=-0.075$). High-pleasure moments (bright yellow) appear at various wanting–liking combinations.

The scatter plot reveals that high pleasure (bright yellow points) occurs both when wanting is high/liking is moderate (upper-left region) and when liking is high/wanting is moderate (lower-right region). This confirms the two-pathway nature of reward in this piece.

8 Temporal Phase Analysis

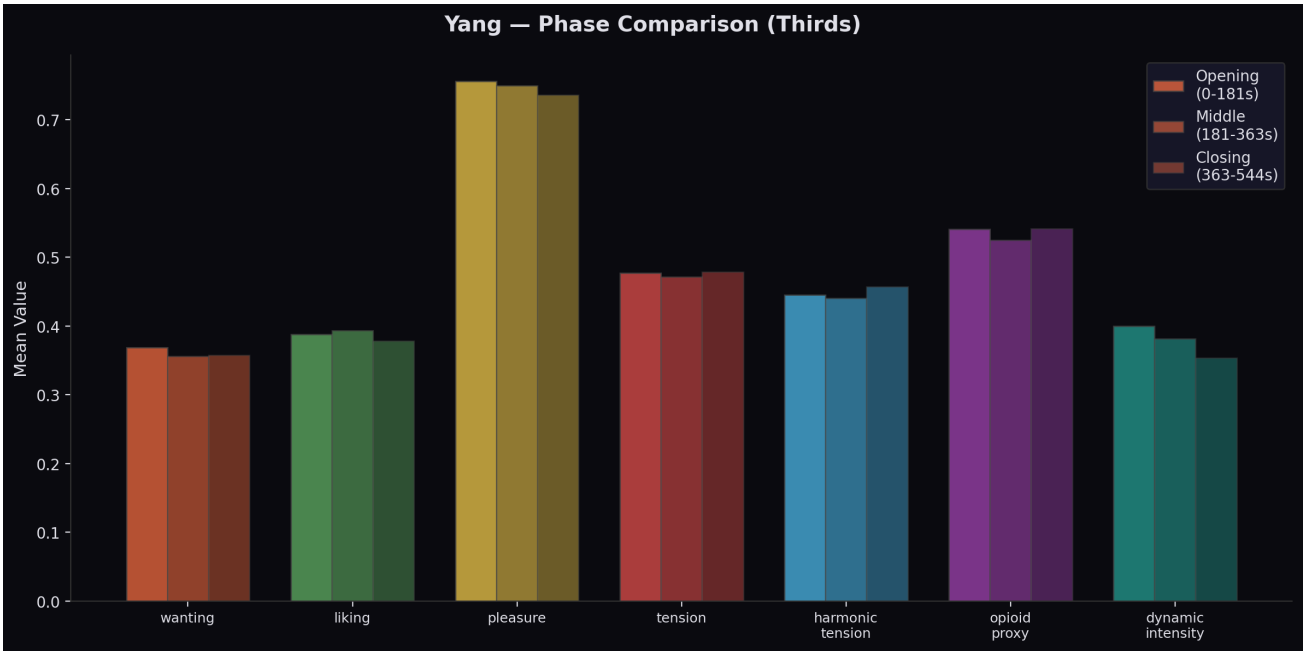

Figure 6: Mean values of 7 key dimensions compared across three temporal phases (thirds). The darkest bars represent the opening, medium the middle, and lightest the closing phase.

Table 6: Phase Analysis — Mean Values by Temporal Third

Dimension	**Opening** (0–181s)	**Middle** (181–363s)	**Closing** (363–544s)	**Trajectory**
wanting	0.369	0.356	0.358	Slight decline
liking	0.388	0.393	0.378	Rise then fall
pleasure	0.756	0.750	0.736	Gradual decline
tension	0.477	0.472	0.479	Stable
harmonic_tension	0.446	0.441	0.458	Gradual rise
opioid_proxy	0.541	0.525	0.542	U-curve
dynamic_intensity	0.400	0.382	0.354	Steady decline

8.1 Phase Trajectory Summary

1. **Opening (0–181s)**: Highest wanting (0.369), highest dynamic intensity (0.400), highest pleasure (0.756). The model finds the opening the most dynamically active and anticipation-driven.
2. **Middle (181–363s)**: Highest liking (0.393), lowest opioid proxy (0.525), lowest tension (0.472). The middle section shows the most consummatory reward and the least tension—a relative plateau.
3. **Closing (363–544s)**: Highest harmonic tension (0.458), highest overall tension (0.479), lowest pleasure (0.736), lowest dynamic intensity (0.354). The model detects increasing harmonic tension with decreasing dynamics toward the end—the piece becomes more harmonically tense while growing quieter.

The overall arc according to the model: the piece opens with the most active reward-seeking and dynamic

activity, settles into a consummatory plateau in the middle, and closes with rising harmonic tension against fading dynamics—**the model detects no conventional resolution**.

9 Comparison with Other Analyzed Pieces

The SRP model has been applied to six pieces. Yang's profile can be compared:

Table 7: Cross-Piece Comparison (Global Means)

Piece	**Dur.**	**Want.**	**Like.**	**Pleas.**	**Opioid**	**Dyn.I.**	**W→L Lag**
Swan Lake	183s	0.362	0.386	0.748	0.502	0.413	14.5s
Duel of the Fates	177s	0.387	0.393	0.780	0.472	0.505	98.9s
Herald of the Change	301s	0.376	0.381	0.757	0.499	0.445	56.3s
Bach Cello Suite	152s	0.377	0.380	0.757	0.485	0.436	2.2s
Beethoven Pathétique	498s	0.369	0.395	0.764	0.512	0.430	−302.7s
Yang	**544s**	**0.361**	**0.387**	**0.747**	**0.536**	**0.378**	**2.6s**

9.1 Where Yang Stands

- **Highest opioid proxy** (0.536): Yang generates the strongest endogenous opioid signal of all tested pieces. The model interprets the piece as providing sustained, deep hedonic satisfaction.
- **Lowest dynamic intensity** (0.378): Yang operates in the narrowest dynamic range. Expressivity is derived from other parameters.
- **Lowest wanting** (0.361): The model finds the lowest anticipatory drive among all pieces, suggesting the piece does not rely on setting up and fulfilling conventional musical expectations.
- **Valid wanting→liking lag** (2.6s): Despite low wanting amplitude, the temporal ordering is correct and falls within the neuroscientific criterion (2–30s from Salimpoor et al., 2011).
- **Longest duration** (544s): Yang is the longest piece analyzed, yet maintains consistent reward values throughout (low standard deviations across all dimensions).

10 Full SRP Visualization

Figure 7: Complete 6-panel SRP visualization for Yang. Top to bottom: (1) Wanting/Liking/Pleasure, (2) DA Caudate/DA NAcc, (3) Tension/Harmonic Tension/Dynamic Intensity, (4) Prediction Error/Prediction Match, (5) Peak Detection/Chills Proximity, (6) Reward Forecast/Resolution Expectation.

11 Technical Notes

11.1 Model Pipeline

1. **Cochlea**: Audio (44.1 kHz) → 128-band log-mel spectrogram
2. **R^3 Spectral** (49D): Consonance (7D), Energy (5D), Timbre (9D), Change (4D), Interactions (24D)
3. **H^3 Temporal** (2304D): 32 horizons (5.8ms–981s) × 24 morphs × 3 temporal laws
4. **Mechanisms** (90D): AED (Affective Entrainment, 30D), CPD (Chills/Peak Detection, 30D), C0P (Cognitive Projection, 30D)
5. **SRP** (19D): Striatal reward pathway simulation using equations from neuroscience

11.2 Key Equations

The SRP wanting and liking are computed as:

$$\text{anticipation} = \sigma\Big(0.5\,(E_{\text{future,15s}} - E_{\text{current}}) + 0.5\,(E_{\text{future,5s}} - E_{\text{current}})\Big) \tag{1}$$

$$\text{DA}_{\text{caudate}} = 0.6 \cdot \text{anticipation} + 0.3 \cdot \text{cpd}_{\text{climax}} + 0.1 \cdot \text{aed}_{\text{expect}} \tag{2}$$

$$\text{DA}_{\text{NAcc}} = 0.6 \cdot \text{c0p}_{\text{cognitive}} + 0.3 \cdot \text{cpd}_{\text{release}} + 0.1 \cdot \text{aed}_{\text{arousal}} \tag{3}$$

$$\text{wanting} = \min(1,\ \beta_{\text{Caudate}} \cdot \text{DA}_{\text{caudate}}), \quad \beta_{\text{Caudate}} = 0.71 \tag{4}$$

$$\text{liking} = \min(1,\ \beta_{\text{NAcc}} \cdot \text{DA}_{\text{NAcc}}), \quad \beta_{\text{NAcc}} = 0.84 \tag{5}$$

where β coefficients are from Salimpoor et al. (2011, Nature Neuroscience).

11.3 Reproducibility

This analysis is fully deterministic and reproducible. The same audio file processed through the same MI pipeline will produce identical results. No random seeds, no stochastic components, no learned parameters.

Generated by Musical Intelligence (MI) — S^3 Spectral Sound Space
Zero learned parameters. Deterministic. Reproducible.